LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART IV

SESSION 13

Climate & Energy-Transition Economics



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Stern, The Economics of Climate Change · KPMG / IFRS Foundation guides on ISSB / TCFD ·
Facilitator: Bongo Adi & Kelechi Nnamdi

RECAP

Part III in Three Ideas

The trilemma

No country can have a fixed exchange rate, free capital flows and independent monetary policy all at once.

Trade follows gravity

Countries trade most with big, nearby partners; AfCFTA aims to lower the barriers that keep intra-African trade small.

Industrial policy is back

Governments now shape supply chains openly — firms must plan for de-risking, friend-shoring and export controls.

TODAY'S PLAN

Session 13 at a Glance — 75 Minutes

Climate change is an economic problem with a price tag — and that price is starting to reach Nigerian balance sheets.

10'	25'	20'	20'

Recap & Hook

Same carbon price, two very different firms

The Economics of Carbon

Externality, carbon price, border adjustments

Exposure & Disclosure

Who pays, CBAM, TCFD/ISSB, stranded assets

Lab: Carbon Stress Test

Your sector at US$25, 50 and 100 a tonne

BY THE END OF TODAY

Session 13 Learning Objectives

You will be able to:

01

Explain Carbon Pricing

Explain why emissions are an externality, and how carbon taxes, trading schemes and border adjustments put a price on them.

02

Size the Exposure

Turn emissions and a carbon price into a naira cost, and compare it with EBITDA by sector and by firm.

03

Plan the Response

Use pass-through, abatement and disclosure to build a credible transition plan for a Nigerian business.

THE HOOK

Same Price, Different Fates



“Carbon pricing is a European problem. We sell in Nigeria.”

Three reasons a Nigerian board should still care:

Exports

The EU's Carbon Border Adjustment Mechanism (CBAM) will charge importers of cement, steel, aluminium and fertiliser for embedded carbon.

Capital

Lenders and investors increasingly price climate risk into the cost of funds and ask for disclosure.

Home policy

Nigeria's Climate Change Act (2021) commits the country to net zero by 2060 and provides for carbon budgets.

Cement vs Solar Under a Carbon Price

The situation

A board is comparing two holdings: a **cement** producer with coal- and gas-fired kilns and a **solar** installer. Both are profitable today.

The decision question

If a carbon price of **US$25, US$50 or US$100 a tonne** applied, how much EBITDA would each lose — and what can management do about it?

The data we have

firm_carbon_exposure.csv — 40 fictional Nigerian listed firms in 9 sectors: revenue, EBITDA, scope 1 and 2 emissions, EU export share, TCFD disclosure. *Stylised teaching data.* We assume ₦1,500 per US$.

What a good answer looks like

A **carbon stress test** by sector and firm, plus a response plan: pass-through, abatement and disclosure.

PART

01

Why Carbon Needs a Price

The biggest externality of all.

Emissions Are an Externality

≋ IN PLAIN ENGLISH

When a kiln or generator emits CO_2, the damage — floods, heat, crop losses — falls on **other people, mostly in the future**. The emitter does not pay, so the market produces **too much** pollution.

EXAMPLE

Nigeria's 2022 floods displaced over a million people. No cement plant, power station or car owner paid for its share of the emissions that make such events more likely.

THE FORMULA

social cost = private cost + damage to others

WHY A MANAGER CARES

When governments correct an externality, costs that were hidden on someone else's books move on to yours. Planning for that is risk management, not politics.

Carbon Tax or Cap-and-Trade?

Carbon tax

Government fixes the price

- Certain price, uncertain emissions
- Simple to run through the tax system
- Revenue can fund transfers or cut other taxes
- Examples: South Africa, Singapore

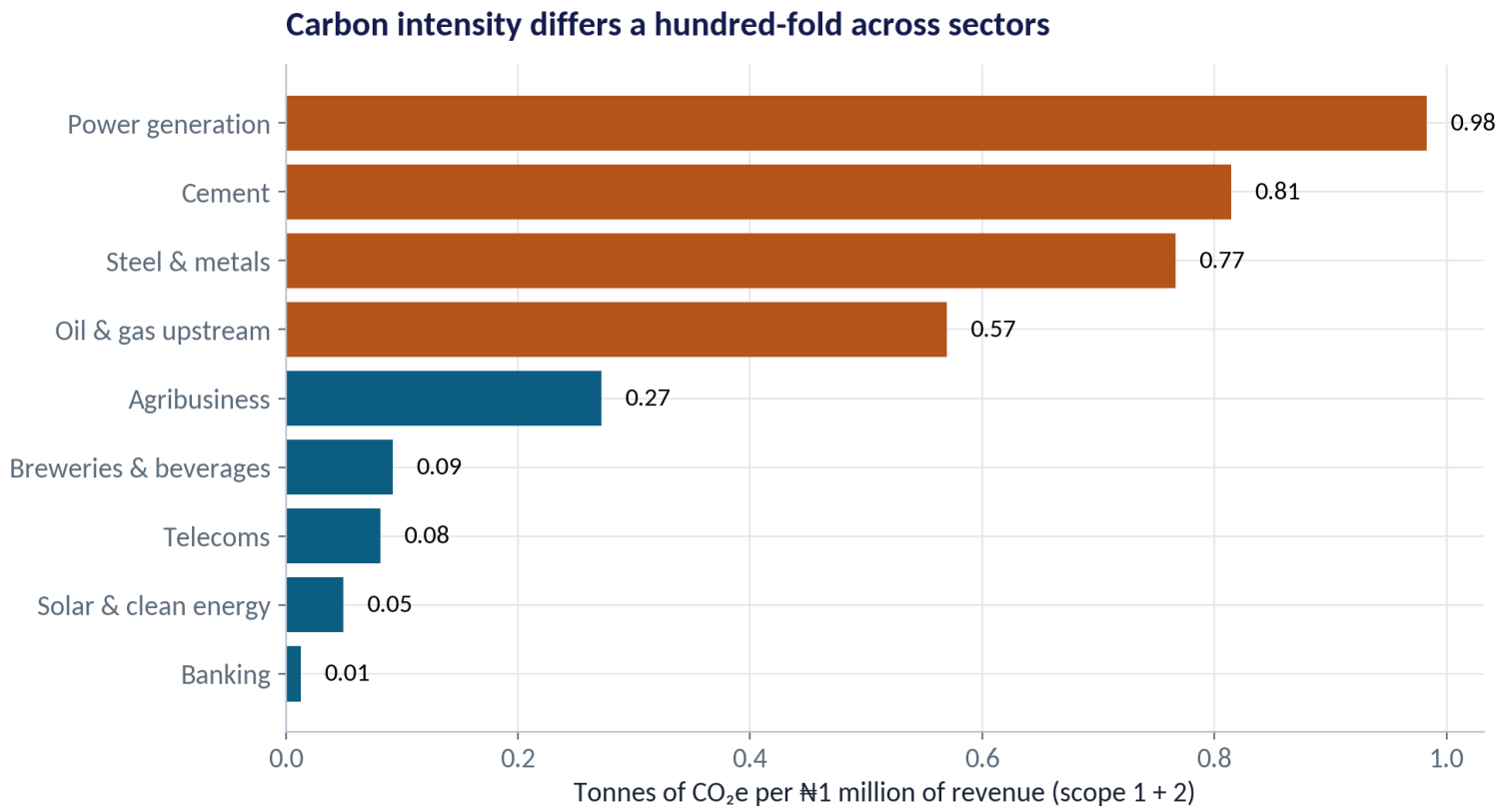
Cap-and-trade (ETS)

Government fixes the quantity

- Certain emissions, uncertain price
- Firms trade permits: cheapest cuts happen first
- Needs strong monitoring and a market
- Examples: EU ETS, China's national ETS

Either way, the firm faces a price per tonne. Our stress test works for both.

Carbon Intensity Differs Enormously by Sector



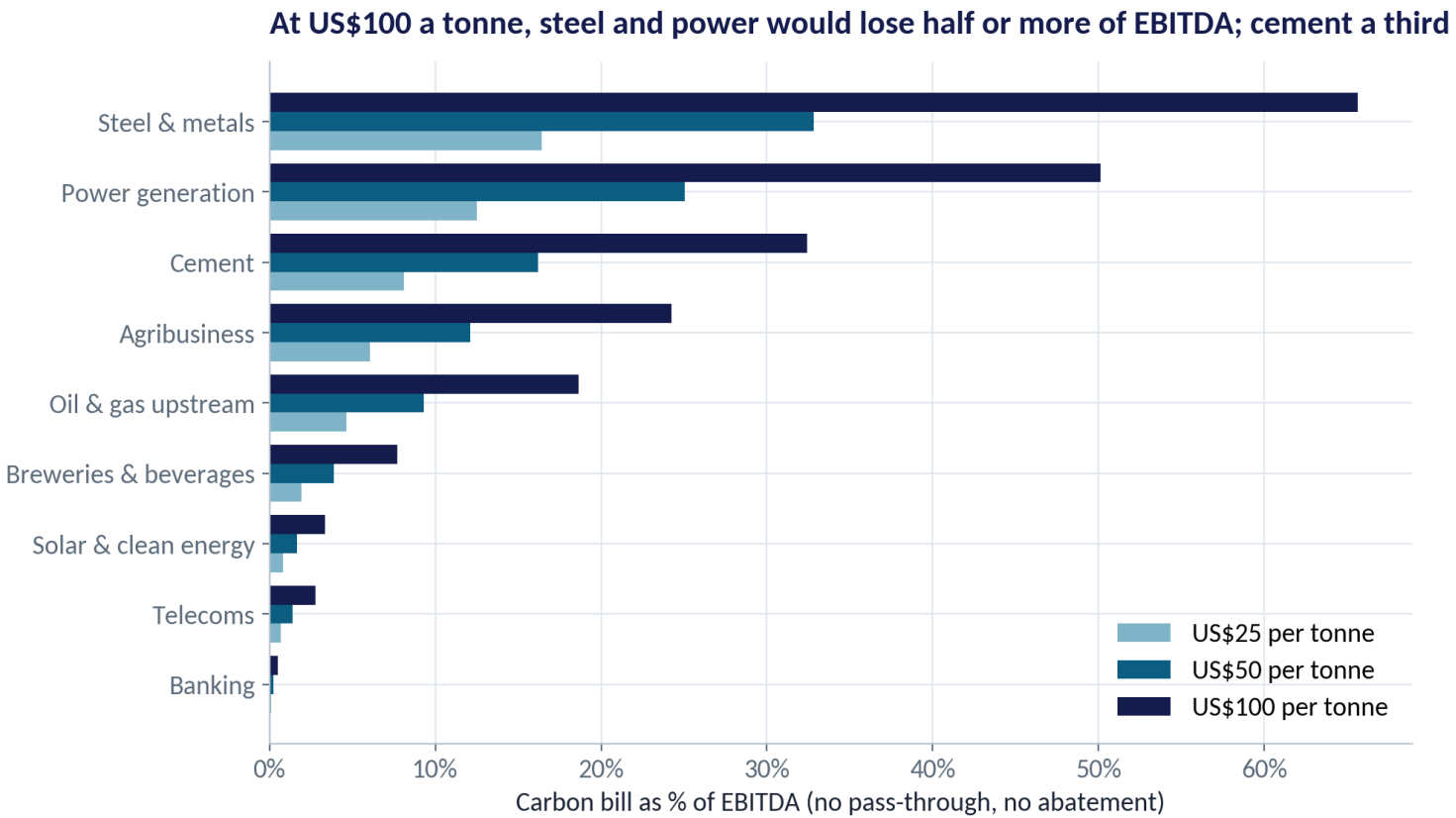
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S13-F01 · replicate: s13_figures.R / .py

WHAT TO SEE

- **Power generation (0.98)** and **cement (0.81)** emit most per ₦1 million of revenue.
- Steel (0.77) and upstream oil & gas (0.57) follow.
- **Banking (0.01)** and telecoms (0.08) emit very little directly — but they lend to and serve the heavy emitters.
- Intensity, not total emissions, tells you how a carbon price hits margins.

The Carbon Bill as a Share of EBITDA



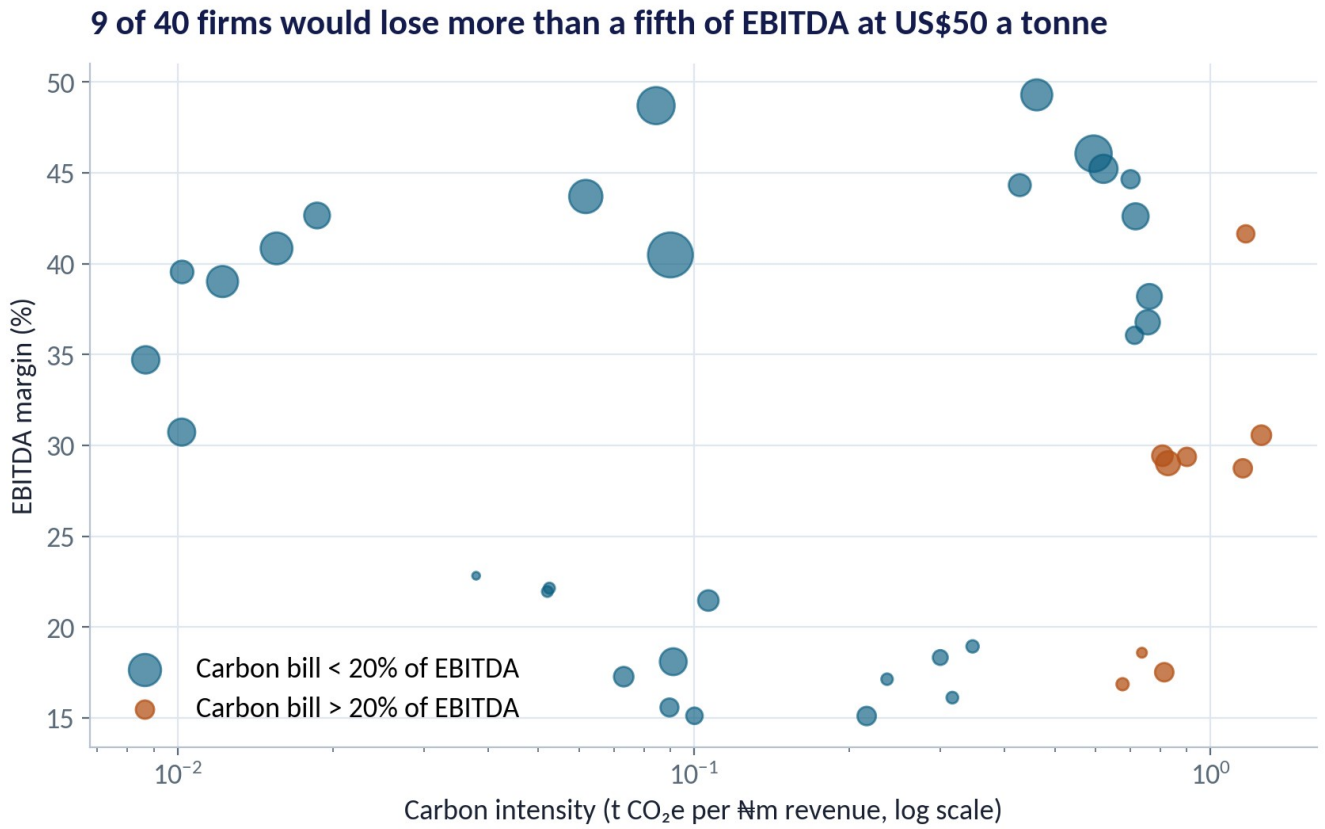
Assumes ₦1,500/US$. Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S13-F02 · replicate: s13_figures.R / .py

WHAT TO SEE

- At **US$50/t**: steel loses about **33%** of EBITDA, power **25%**, cement **16%**.
- At **US$100/t**: steel **66%**, power **50%**, cement **32%**.
- Banking, telecoms and solar lose **under 3%** even at US$100.
- These are worst cases: no pass-through and no abatement.

Who Is Really Exposed?



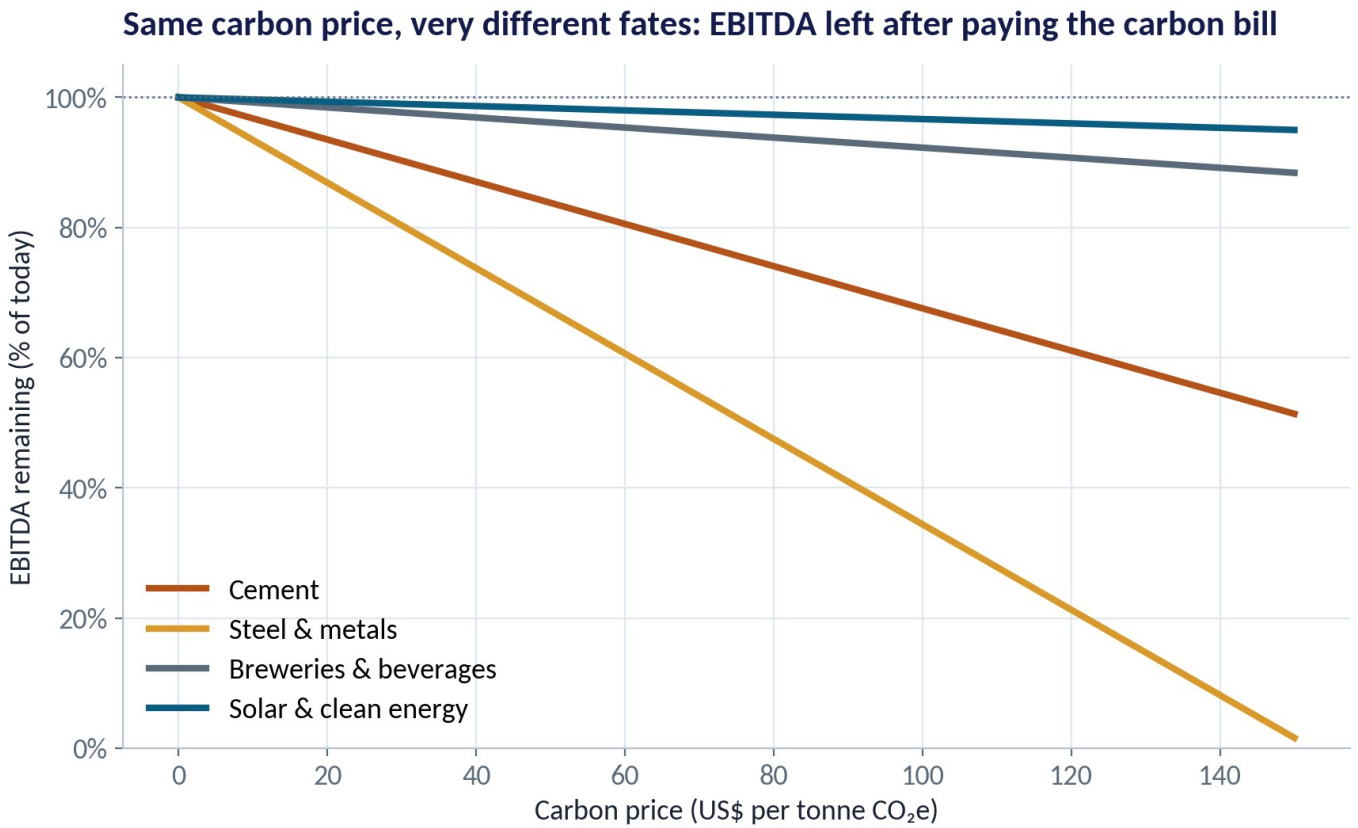
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S13-F03 · replicate: s13_figures.R / .py

WHAT TO SEE

- High intensity **and** a thin margin is the danger zone.
- **9 of 40** firms would lose more than a fifth of EBITDA at US$50/t.
- Firms within one sector differ — the best cement plant is far cleaner than the worst.
- Investors will reward the clean firms within dirty sectors.

Cement vs Solar as the Carbon Price Rises



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S13-F04 · replicate: s13_figures.R / .py

WHAT TO SEE

- Cement EBITDA falls steadily as the price rises — about **16%** lower at US$50.
- Solar is almost unaffected — and may gain if it sells power to emitters.
- The same policy **moves value** from high-carbon to low-carbon businesses.
- This is what investors mean by **transition risk**.

CONCEPT CHECK

Quick Check: What Would You Say?

?

At US$50 a tonne, the stylised cement sector's carbon bill is about ₦252bn, or 16% of its EBITDA. A director says: 'That is only 6% of revenue — not a big deal.' What is the best reply?

A. Agreed — what matters is revenue

B. The relevant comparison is profit: losing a sixth of EBITDA, before any pass-through, is material for valuation and debt covenants

C. The bill is zero because Nigeria has no carbon price

D. Carbon costs are always passed on fully to customers

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Borders, Prices and Disclosure

How carbon policy abroad reaches Nigerian firms.

The EU's Carbon Border Adjustment Mechanism

IN PLAIN ENGLISH

CBAM makes EU importers pay for the carbon embedded in certain goods — cement, iron and steel, aluminium, fertilisers, electricity and hydrogen — at roughly the EU carbon price, minus any carbon price already paid where the goods were made.

EXAMPLE

For the stylised cement and steel exporters, an EU price of **US$80/t** costs about **8–14% of EU export revenue** — but only **2–11% of total EBITDA**, because most sales are domestic.

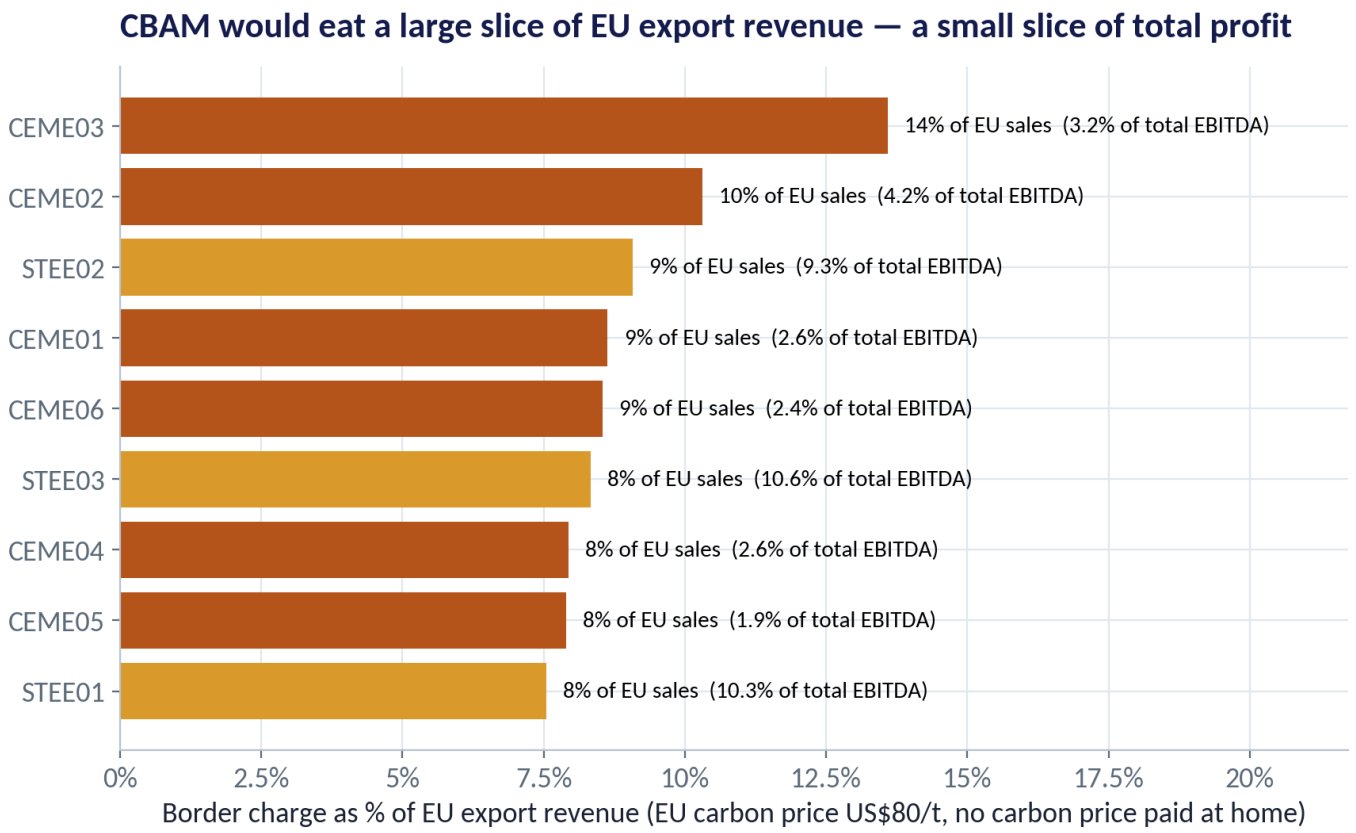
THE FORMULA

CBAM charge ≈ embedded tonnes × (EU price − carbon price paid at home)

WHY A MANAGER CARES

Reporting started in October 2023; payment obligations begin for imports from 2026. Exporters that measure and cut their emissions keep their EU customers.

Big Slice of EU Sales, Small Slice of Profit



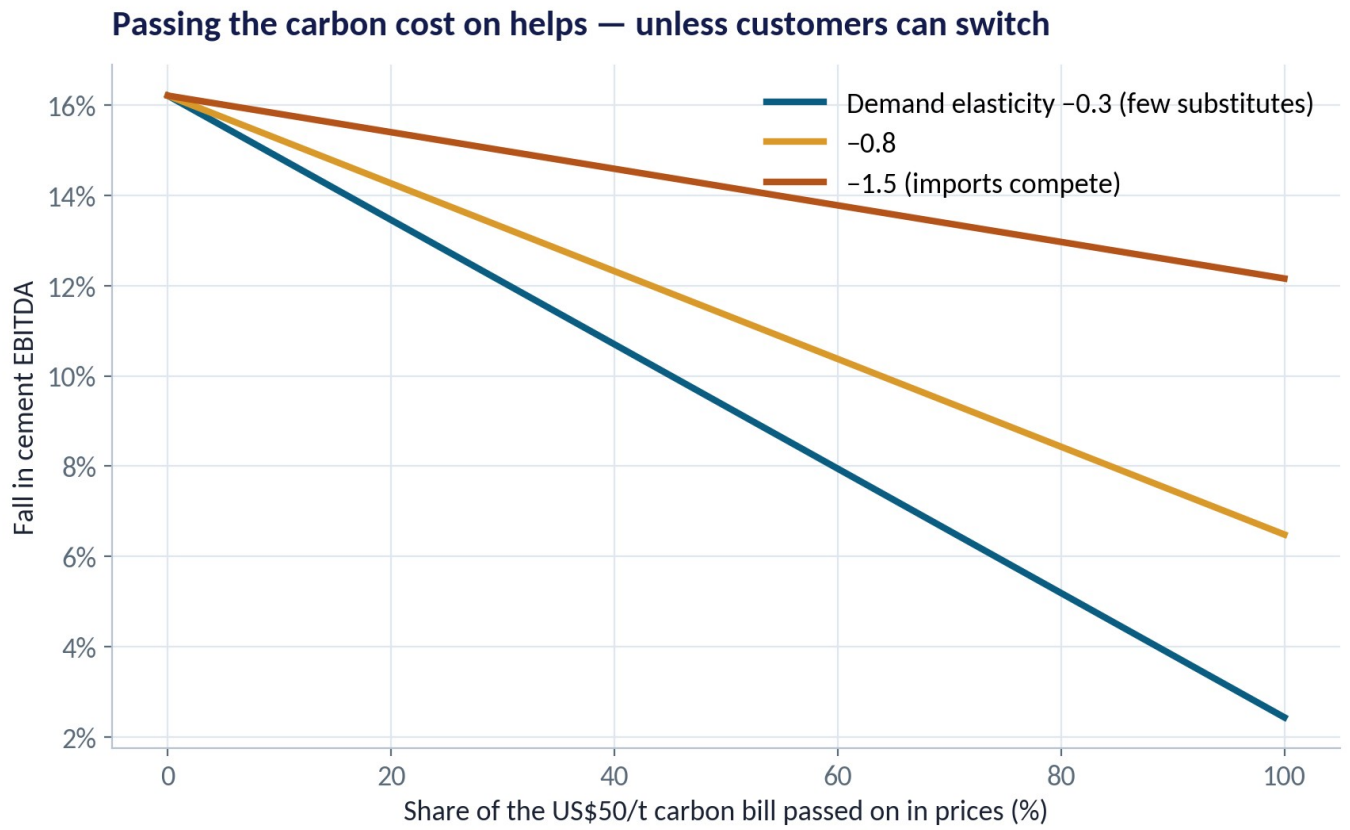
Rust = cement, gold = steel. Embedded emissions approximated by EU export share × scope 1. EEGB stylised data.

Fig S13-F05 · replicate: s13_figures.R / .py

WHAT TO SEE

- Cement and steel are the Nigerian sectors most directly covered.
- The charge equals **8–14%** of EU export revenue for these firms.
- Because EU sales are a small share of total sales, the hit to group EBITDA is smaller.
- But the EU price is the **benchmark** other markets may follow.

Can Cement Pass the Cost On?



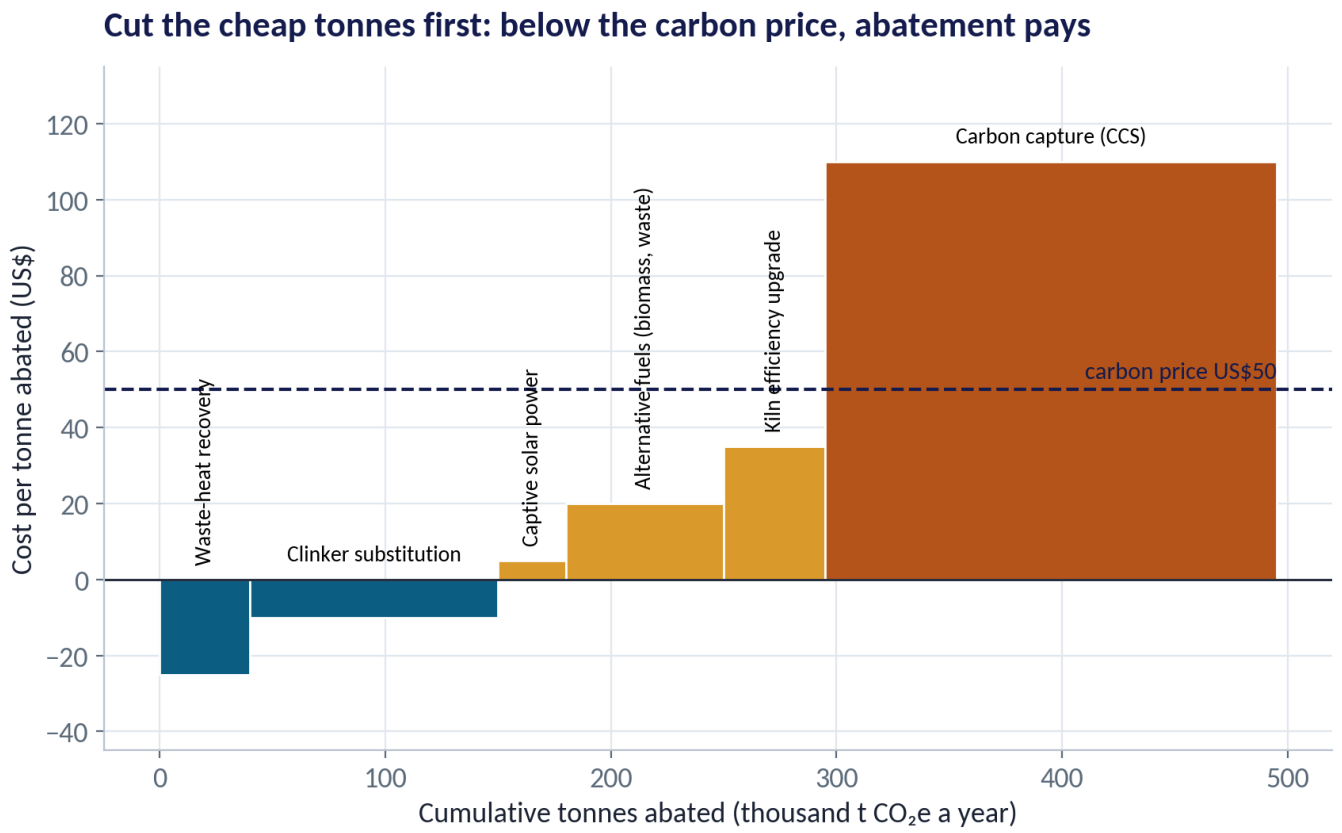
Illustrative: 50% contribution margin on lost volume. EEGB stylised teaching data.

Fig S13-F07 · replicate: s13_figures.R / .py

WHAT TO SEE

- Passing the bill on in prices protects margin — if customers stay.
- With few substitutes (elasticity **−0.3**), full pass-through leaves EBITDA only about **2%** lower.
- If imports compete (elasticity **−1.5**), lost volume still costs about **12%** of EBITDA.
- Session 2's elasticity is the key to any carbon strategy.

Cut the Cheap Tonnes First



Illustrative abatement options for one 700 kt cement plant; costs are teaching assumptions, not engineering estimates.

Fig S13-F08 · replicate: s13_figures.R / .py

WHAT TO SEE

- A marginal abatement cost curve ranks options by **cost per tonne saved**.
- Waste-heat recovery and clinker substitution **save money** even with no carbon price.
- Below a US$50 price, about **295 of 495** thousand tonnes can be cut profitably.
- Carbon capture is the expensive last resort.

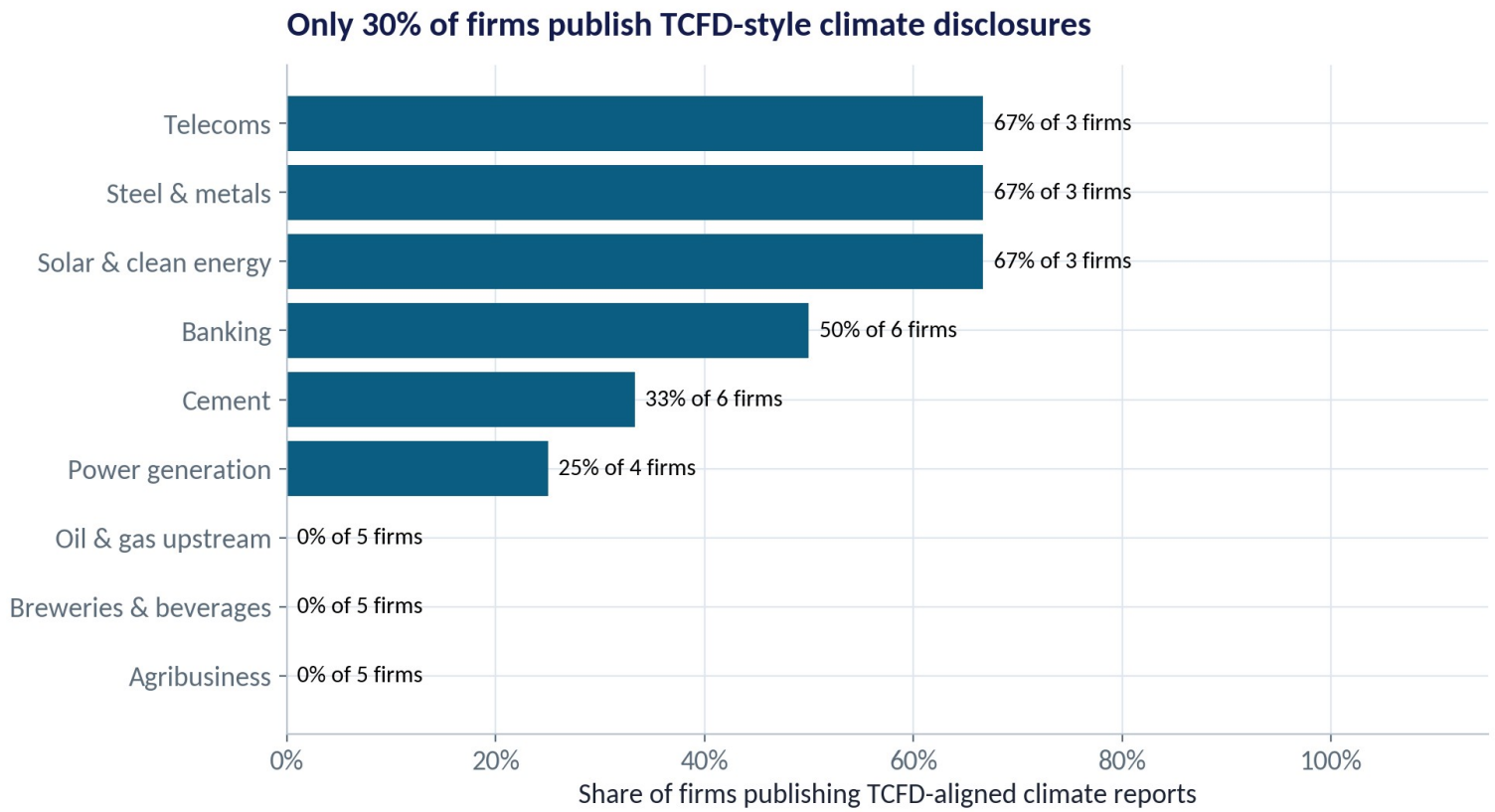
DISCLOSURE

From TCFD to ISSB: What Investors Will Ask

Pillar	The question	What a Nigerian firm should show
Governance	Who oversees climate risk?	Board committee and management owner named
Strategy	How could climate change hurt or help us?	Scenarios such as US$25 / 50 / 100 per tonne; physical risks (floods, heat)
Risk management	How do we find and manage these risks?	Climate in the risk register, with owners and limits
Metrics & targets	How much do we emit and what is the plan?	Scope 1 and 2 (later 3), intensity, and dated reduction targets

The ISSB's climate standard (IFRS S2, 2023) builds on these four TCFD pillars. Nigeria's Financial Reporting Council has set out a roadmap to adopt ISSB standards.

Only a Minority Report Today



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S13-F06 · replicate: s13_figures.R / .py

WHAT TO SEE

- About **30%** of the stylised firms publish TCFD-style reports.
- Disclosure is highest in telecoms, solar and steel (2 of 3 firms each).
- None of the upstream oil & gas firms disclose — exactly where investors most need data.
- Early movers gain credibility with lenders and DFIs.

PART

03

Africa's Transition Story

Risk, but also opportunity.

Where the Opportunities Are

Energy access

Tens of millions of Nigerians lack reliable grid power. Solar mini-grids and home systems replace costly diesel and petrol generators.

Critical minerals

Lithium, cobalt and other minerals in Africa feed global battery supply chains. Value lies in processing, not only mining.

Gas as a bridge

Nigeria's Energy Transition Plan treats gas as a transition fuel. Firms must plan for when that bridge ends.

Adaptation & green finance

Green bonds, adaptation funds and DFI lending lower the cost of capital for credible transition projects.

How Boards Get Climate Risk Wrong

THE TRAP	THE BETTER HABIT
✗ “There is no carbon price in Nigeria, so it doesn't affect us.”	✓ Run **scenarios**: CBAM, lender requirements and future domestic policy.
✗ Measuring only total emissions.	✓ Track **intensity** per unit of output and revenue — that drives margins.
✗ Assuming every carbon cost can be passed on.	✓ Check **elasticity and import competition** first.
✗ Treating disclosure as a PR exercise.	✓ Use TCFD/ISSB as a **management tool**: owners, metrics, targets.
✗ Ignoring physical risk.	✓ Floods and heat already disrupt plants, roads and harvests in Nigeria.

Quick Check: What Would You Say?

?

A cement firm can cut 110 thousand tonnes of CO_2 by substituting clinker, at a cost of –US$10 a tonne (it saves money). Why might it not have done so already?

- **A** Because the option saves money, it must already have been done
- **B** Upfront capital, management attention, technical risk or customer acceptance can block profitable cuts — a carbon price or disclosure pressure tips the balance
- **C** Abatement only makes sense when the carbon price is above US$100
- **D** Clinker substitution increases emissions

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

HANDS-ON LAB

Lab: Carbon Stress Test for Your Sector

YOU WILL NEED

Data
`data/firm_carbon_exposure.csv`

R script
`sessions/s13/R/s13_lab.R`

Python script
`sessions/s13/python/s13_lab.py`

⏱ **20 minutes**

1. Compute each firm's **total emissions** (scope 1 + 2) and **intensity** per ₦1 million of revenue.
2. Compute the **carbon bill** at US$25, 50 and 100 a tonne (₦1,500/US$).
3. Express it as a **% of EBITDA** by sector and rank the sectors.
4. List the **ten most exposed firms** and check whether they disclose.
5. Rerun with a different exchange rate and with a 30% abatement.

Deliverable: three sentences for a cement board: our exposure, what customers will bear, and our first two actions.

Today's Vocabulary in Plain English

Externality	A cost or benefit that falls on people outside the transaction.
Carbon price	A charge per tonne of CO_2 emitted, via a tax or permits.
ETS	Emissions trading system: a cap on emissions with tradable permits.
CBAM	The EU's charge on embedded carbon in certain imports.
Scope 1 / 2	Direct emissions / emissions from purchased electricity.
Carbon intensity	Emissions per unit of output or revenue.
Transition risk	Losses from policy, technology and market shifts to low carbon.
Physical risk	Losses from floods, heat, drought and storms.
Stranded asset	An asset that loses value early because of the transition.
MACC	Marginal abatement cost curve: emission cuts ranked by cost per tonne.

KEY TAKEAWAYS

Session 13 in 5 Ideas

01 Emissions are an **externality**; a carbon price makes the emitter pay.

02 Exposure depends on **intensity and margins**, not on total emissions alone.

03 **CBAM** brings EU carbon prices to Nigerian cement and steel exports.

04 Pass-through depends on **elasticity**; abatement should start with the **cheapest tonnes**.

05 Disclosure under **TCFD/ISSB** is becoming the price of access to capital.



NEXT › Session 14: Country Risk & Scenarios

Before next session: Read India: Liberalise in the Face of Crisis? Note the warning signs before 1991 and how policymakers responded. Bring one indicator you would watch to judge a country's risk.